

Research article

Do environmental fluctuations during development affect trait variation? An experimental test with salinity

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Human-induced climate change is a driver of extinction, with extreme events occurring more frequently. It increases both the amount and fluctuations in environmental stress that organisms experience. In such environments, greater intra-specific trait variation creates more potential for adaptation through natural selection. Many studies focus on changes in trait means between novel and historic environments but overlook changes in trait variation. Here we tested how salinisation – which currently affects 20–50% of freshwaters worldwide – alters trait variation in a freshwater fish, *Gambusia holbrooki*. We reared newborn fish in freshwater, stable-salinity or fluctuating-salinity water until maturation, and then compared variation in fitness-related traits in each sex during early and late adulthood. Salinisation had stronger effects on young virgin adults than older mated adults, with sex-specific impacts (female: gut development; male: age at maturity and body size). When we accounted for the mean trait value in each environment, salinisation also affected relative variation (i.e. the coefficient of variation) in female fecundity (egg size, offspring number). Notably, the fluctuating-salinity treatment did not magnify the effect of the stable-salinity treatment on trait variation but sometimes reversed its effect. Our findings suggest that researchers should pay closer attention to fluctuating environmental stressors, as they can differ from stable stressors in how they affect between-individual variation in trait expression.

Keywords: climate variability, intraspecific trait variation, life history, salinisation, sex difference

Introduction

Extreme events are now occurring more frequently globally (Harvey et al. 2022, Raynal et al. 2022), with human-induced climate change affecting both the extent and frequency of environmental stress experienced by organisms, such as heavy rainfall (Li and Otto 2022), heatwaves (Woolway et al. 2022) and aridity (Wang et al. 2022). Fluctuations in environment activate numerous physiological

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pathways that animals use to maintain homeostasis (Evans and Kültz 2020). Such fluctuations experienced early in life can expose hidden genetic variation, reduce canalisation and increase trait variation in a population (Rowiński and Rogell 2017). Increased trait variation in a population can also result from greater individual differences in how resources are allocated when animals are stressed (Dupont-Prinet et al. 2010), leading to higher variation in life-history strategies. Greater trait variation under current environmental conditions implies that individuals with traits that deviate from the norm have reduced survival capacity (Speakman et al. 2004). However, the fitness consequences of greater trait variation can be context-dependent (Burton et al. 2011). Trait values that were common in the original environment are often sub-optimal in novel environments (Ghalambor et al. 2007), where increased trait variation creates a greater opportunity for selection: the subset of individuals with fortuitously advantageous trait expression are more likely to thrive (Bolnick et al. 2011). This leads to trait evolution if there is genetic variation in developmental responses to novel environments (Ledón-Rettig et al. 2010, McGuigan et al. 2011).

Males and females often differ in how trait expression changes in a more challenging environment (Harding et al. 2019). This difference is partly due to sex-specific selection on allocation to reproduction and somatic maintenance (Teder and Kaasik 2023). Sexual selection is typically stronger on males (Janicke et al. 2016), favouring greater investment in traits that increase mating and fertilization success, thereby diverting resources from self-maintenance (Chung et al. 2024a). In contrast, females rarely struggle to acquire a mate and often benefit from greater allocation to self-maintenance, which enhances fecundity (Bonduriansky et al. 2008). Consequently, males seem to be more sensitive to environmental change (Mauvais-Jarvis 2024) and may exhibit greater variation in fitness-related traits when stressed (Martinossi-Alilibert et al. 2017). However, the magnitude and direction of sex differences in trait variation can also depend on the trait's function. For example, males usually have greater variation than females in sexually selected traits (Wyman and Rowe 2014), whereas females show greater variation than males in immunocompetence (a key life-history trait that affects survival by defending against pathogens) (Zajitschek et al. 2020). Increased variation in male sexual traits can accelerate selection on males when females choose males based on sexual traits that signal offspring fitness (Jennions and Perrie 1997). On the other hand, greater variation in female life-history traits implies variation in their performance of survival-enhancing tasks (e.g. parasite resistance). Greater variation potentially lowers the risk of extinction when the environment fluctuates because at least some individuals will still survive and breed (Bolnick et al. 2011). To understand how population extinction risk changes when individuals are exposed to greater stress, and whether this risk is modified by stressors being stable or fluctuating, it is essential to test for sex differences in trait variation in different types of environments.

Here, we test how trait variation in males and females shift in response to changes in salinity during development in a freshwater fish. Salinity levels fluctuate naturally due to precipitation, evaporation, and events such as flooding and the isolation of water pools (Van De Pol et al. 2010, Cañedo-Argüelles et al. 2019). For example, the weathering of aquifer minerals, followed by the dissolution of salt deposits, leads to freshwater salinisation in arid and semi-arid zones (Jalali 2007). Likewise, meltwater after snowfalls can introduce minerals and nutrients from soil into freshwater systems, which increases salinity (Beegum et al. 2023). However, these natural fluctuations are now exacerbated by human activities (e.g. irrigation, mining, road de-icing) (Iglesias 2020, Cunillera-Montcusí et al. 2022). For example, wastewater from shale gas operations has caused salinity levels to become seven times higher than that of seawater in some places (Vengosh et al. 2014). Also, millions of tons of road salt are applied during winter for de-icing (Hintz et al. 2022), subsequently being washed into nearby freshwater systems (Corsi et al. 2010). As a result, salinisation has impacted 20–50% of freshwater globally (Schuler et al. 2019).

In freshwater, fish are hyper-osmotic relative to their surroundings. To maintain osmotic balance, freshwater fish excrete excess water as dilute urine and actively absorb salts via ion-transporters in their gill and intestines in freshwater (Kültz 2015). In contrast, marine fish drink water, excrete excess ions through the gill, and reabsorb water via the intestines to counter water loss in high-salinity environments (Lavery and Skadhauge 2012). These processes require the activation and expression of osmoregulatory enzymes in mitochondria-rich cells (Hwang and Lee 2007), which are often associated with elevated levels of stress hormone (McCormick 2001) and increased metabolic rates (Zikos et al. 2014). Freshwater fish may achieve osmotic balance after long-term acclimation to a stable saline environment, but frequent fluctuations in salinity can destroy the balance (Evans and Kültz 2020). Osmotic imbalance may amplify energy costs due to the need to switch between different osmoregulatory modes (Kültz 2015), potentially diverting resources from other functions, such as reproduction and growth (Neves et al. 2019). Most freshwater fish species inhabit environments with relatively stable salinity and struggle to acclimate to rapid changes in salinity (Edwards and Marshall 2012). This underscores the direct threats that salinity fluctuations pose for freshwater fish (Evans and Kültz 2020). Despite the crucial role of fish in freshwater ecosystems (Villéger et al. 2017, Toussaint et al. 2018), only 16 of 193 recent studies on the effects of freshwater salinisation were on fish (reviewed by Cunillera-Montcusí et al. 2022), and few (if any) of these studies investigated how trait variation changes when salinity fluctuates.

To test the effect of salinity during development on trait variation between individuals, we ran an experiment using the mosquitofish, *Gambusia holbrooki* (Fig. 1) – an invasive freshwater fish with a global distribution (Pyke 2008). Many established populations have ready access to saline waters in the wild (Morgan et al. 2003, Alcaraz and García-Berthou



Figure 1. A male (lower) and a female (upper) *Gambusia holbrooki*. Photographed by Andrew T. Kahn.

2007, Ruiz-Navarro et al. 2011, Riesch et al. 2015), where they compete with endemic fish and amphibian species (Alcaraz et al. 2008, Karraker et al. 2010). Field studies have reported life-history changes in these wild *Gambusia* populations with an increase in water salinity, including faster ovarian development (Brown-Peterson and Peterson 1990, Alcaraz and García-Berthou 2007, Martin et al. 2009), poorer offspring quality (Santi et al. 2020), shorter recruitment periods (Ruiz-Navarro et al. 2011) and changes in somatic condition – either an improvement (Ruiz-Navarro et al. 2011, Riesch et al. 2015) or a decline (Alcaraz and García-Berthou 2007, Martin et al. 2009). Disentangling the effects of salinity and its fluctuations from these studies is challenging. The level of salinity fluctuations is often unknown (Stearns and Sage 1980, Alcaraz and García-Berthou 2007) and there are confounding environmental factors (e.g. food availability, predation risk) that covary with salinity gradients (Ruiz-Navarro et al. 2011). In contrast, experimental studies that transferred *Gambusia* from freshwater to high-salinity environments often revealed a negative effect: reduced foraging effort (Alcaraz et al. 2008), lower aggression (Alcaraz et al. 2008), decreased exploratory behaviour (Zhou et al. 2022) and increased expression of stress-related enzymes (Tsai et al. 2018). However, these studies have primarily focused on adult fish exposed to salinity environments, limiting our understanding of how early-life salinity exposure influences trait variation.

In general, a salinity level of approximately 10‰ is considered to be an eco-physiological barrier that separates freshwater and marine fauna (Telesh and Khlebovich 2010). However, current salinisation often causes the salinity levels to approach or exceed this threshold, such as 8‰ due to tidal flow (Brown-Peterson and Peterson 1990), 13.5‰ due to road salt runoff (Hintz and Relyea 2019), 19‰ due to land clearing (Ingersoll et al. 1992, Cartwright et al. 2007), 30‰

due to potash mining (Coring and Bätke 2011) and 35‰ due to irrigation (Velasco et al. 2006). These elevated levels fluctuate due to variable climate events (e.g. flooding and drought) and temporary contaminant runoff (e.g. seasonal use of road salt). The changes in salinity levels are expected to be more severe in shallow ponds with low water volume (Brady and Benoit 2025) – a common habitat for *Gambusia* (Pyke 2005).

To test how the critical salinity threshold (10‰) affects trait variation, we reared *G. holbrooki* from birth to maturity in either freshwater (0‰), stable salinity (10‰) or fluctuating salinity (0–20‰; mean = 10‰). We measured key life-history traits (size and age at maturity, relative gut length, immunity) in both sexes and sex-specific traits (size of intermittent organ, sperm/egg production, total offspring number). These traits were measured twice (in early and late adulthood) to test for age-dependent effects. In our previous work on changes in trait means, we found that stable salinity led to earlier maturation in both sexes, greater offspring number, larger female size, but reduced female gut development; in contrast, fluctuating salinity delayed female maturation, reduced reproductive output in both sexes and caused smaller male size (Chung et al. 2024b). Nevertheless, the trait mean and trait variation can respond to environmental changes differently, potentially altering the environmental impact on population viability (Zheng et al. 2023). In this study, we investigated how salinisation affected variation in these traits.

Material and methods

Experimental overview

Adult mosquitofish were collected from a freshwater stream (0‰ salinity) in Canberra, Australia (35°16′44.4″S, 149°06′57.6″E) in 2020–2021 and held in 90-l freshwater tanks (ca 50 fish/tank). Wild-caught pregnant females were transferred to individual 1-l freshwater tanks to give birth (occurring 7–30 days after transfer). Newborn fry from 118 broods were randomly assigned to three treatments (maximum 11 fry per brood/treatment): freshwater (0‰; n = 343), stable salinity (10‰; n = 397), or fluctuating salinity (0–20‰ with mean = 10‰; n = 392). Siblings in the same treatment were housed in communal tanks at a density of ≤ 1 individual l⁻¹. Three tanks (one per treatment) for each brood were placed near each other to minimise differences in rearing conditions (e.g. temperature; light) within temperature-controlled rooms. We accounted for potential differences among broods in our analyses. All fish were kept under a 14 light:10 dark cycle at $28 \pm 1^\circ\text{C}$ and fed *Artemia* nauplii twice daily.

We manipulated salinity concentrations by dissolving aquarium salt (Aquaforest, Brzesko, Poland) with aged tap water and aquatic conditioner (Seachem, Madison, USA). We monitored salinity every two days using a refractometer and renewed the water weekly. Fish in the freshwater treatment were kept at 0‰ salinity. To minimise mortality due to initial salinity exposure, newborn fish in the two salinity

treatments were first transferred to tanks with 2.5‰ saline water. After two days, fish in the stable salinity treatment were then kept at 10‰ salinity, while those in the fluctuating treatment were exposed to cyclically varying salinities (10 → 20 → 10 → 0 → 10‰). Fluctuating salinity levels were generated by rotating fish between tanks with different salinity levels every two days. To control for any effect of being transferred, fish in the freshwater and stable-salinity treatments were also rotated between tanks with the appropriate water (0 or 10‰, respectively).

We inspected fish every two days to determine the date of maturation (gravid spot for females; a pointed anal fin for males). Newly matured adults were individually transferred to 1-l tanks of freshwater. After one week in this common garden setting, we euthanised a random subset of females ($n=72$ freshwater; 118 stable salinity; 100 fluctuating salinity) and males ($n=68$ freshwater; 81 stable salinity; 92 fluctuating salinity) to test if the developmental environment affected life-history traits and reproductive traits of young adults. To test the long-term effect of the developmental environment, we paired each of the remaining females ($n=99$ freshwater; 98 stable salinity; 87 fluctuating salinity) and males ($n=104$ freshwater; 100 stable salinity; 113 fluctuating salinity) with a stock fish of the opposite sex in a 4-l freshwater tank to breed. We replaced the stock fish fortnightly to maintain the sexual interests of focal fish. After 12 weeks, focal fish were euthanised to measure life-history and female reproductive traits. Male reproductive traits did not require euthanasia, so we measured these traits on the same males when young and old (i.e. before and after the mating period).

Life-history traits

Body length

Body size is a key determinant of *G. holbrooki* fitness. Larger individuals tend to outcompete smaller counterparts (Matthews and Wong 2015, Harrison et al. 2018), be more sexual attractive (Bisazza et al. 1989) and have higher overwinter survival (Reznick and Braun 1987). Therefore, salinity-induced changes in size variation are likely to influence the strength and outcome of intraspecific competition. We measured body size by anaesthetising fish in ice-water. The fish was placed on a glass slide with a scale, and the side of its body was photographed to measure its standard length (snout tip to end of vertebral column) using a digital camera. The photos were later analysed using ImageJ software ver. 2.1.0.

Immune response

The immune system protects animals against common microorganisms and defends against specific pathogens upon reinfection (Hoebe et al. 2004). Individual variation in immunocompetence affects the susceptibility to parasitism and diseases, as well as influencing resources allocation between immune defences and other functions (McAfee et al. 2021). We used an in vivo reaction to phytohemagglutinin to measure cell-mediated immunity (verified in *G. holbrooki*: Iglesias-Carrasco et al. 2019). Once a fish was anaesthetised, we used a spessimeter (Mitutoyo 547–301; accuracy: 0.01

mm) to measure the thickness of the posterior end of its dorsal fin five times. We then injected 0.01 mg phytohemagglutinin dissolved in 0.01 ml of PBS solution. After 24 h, the thickness was again measured five times. The difference between the pre- and post-injection measures (i.e. swelling) is an index of immunity, which involves non-specific inflammation and immune cell recruitment (Kennedy and Nager 2006).

Relative gut length

Gut length generally correlates with nutrient absorption capacity, with a longer gut often extending the period of food retention (Cant et al. 1996). In addition, fish guts can facilitate ion regulation and water uptake, which are critical to acclimate to salinity (Barany et al. 2020). We measured variation in relative gut length (i.e. accounting for body size: allometric size adjustments) to assess potential changes in digestion and osmoregulation. After the immunity test, we dissected fish to extract and photograph their gut (the digestive tract from mouth to anus). We measured its length using ImageJ.

Female reproductive traits

We dissected females to count how many eggs they held. For young females (still virgins), we photographed their unfertilised eggs and measured the diameter using ImageJ. We did not measure the eggs of old, mated females because they carried embryos (whose size varies with their developmental stages; Riesch et al. 2011). Instead, we recorded a) the number of embryos they held at the end of the experiment and b) the number of offspring they had produced during the mating period. We considered (a) + (b) as the total offspring number.

Male reproductive traits

We measured the length of the male intromittent organ (gonopodium) from photographs. We recorded sperm count and velocity following methods in Chung et al. (2021). A male was anaesthetised in icy water and placed on a glass slide covered with a 1% polyvinyl alcohol solution under a dissecting microscope. We swung his gonopodium forward and gently pressed his abdomen to expel his sperm reserves. The male was then returned to his 1-l tank for sperm replenishment. After seven days, we re-stripped the male to collect the sperm into an Eppendorf tube with 400–1200 µl of extender medium (207 mM NaCl, 5.4 mM KCl, 1.3 mM CaCl₂, 0.49 mM MgCl₂, 0.41 mM MgSO₄, 10 mM Tris (Cl); pH 7.5). We vortexed the sperm solution for 30 s and mixed it using a 10-µl pipette. Three µl of the solution was then placed on a 20-µm capillary slide (Leja) to count the sperm number using a CEROS Sperm Tracker (Hamilton Thorne Research, Beverly, MA, USA) under 100× magnification. The sperm count was calculated using the mean value of five randomly selected subsamples per male (repeatability: $r=0.829 \pm 0.010$ SE, $p=0.001$, $n=525$ ejaculates). We were unable to analyse 28 samples due to the shortage of capillary slides during the Covid-19 lockdown.

For sperm velocity, we collected two separate samples of four sperm bundles into an Eppendorf tube containing 2 μ l of extender medium. Each sample was analysed separately. We placed 3 μ l of the sperm sample into a cell in a 12-cell multi-test slide (MP Biomedicals, USA) coated with 1% polyvinyl alcohol solution, activated the sample using a 3 μ l solution of 125 mM KCl and 2 mg ml⁻¹ bovine serum albumin for 30 s, and then covered it with a coverslip. We measured the actual velocity along the sperm trajectory (i.e. curvilinear velocity) using a CEROS Sperm Tracker. The weighted average of the motile sperm tracks in both samples (66.66 ± 1.45 SE tracks; $n = 553$ ejaculates) was considered as sperm velocity.

Statistical analysis

Analyses were conducted in R studio ver. 1.3.1093 with R ver. 4.0.5 (www.r-project.org). We estimated covariances between traits across individuals using Bayesian multivariate mixed effects models (*brm* function in the 'brms' package ver. 2.22.0 (Bürkner et al. 2024)) with all traits modelled together for each of the four age/sex groups separately. We ran separate models for the life-history and reproductive traits of young males because these were measured on different males. We treated environment (three levels) as a fixed factor and brood identity as a random factor to account for measurement of several fish from the same brood. 'Age at maturity', 'immunity', 'egg number of young females' and 'sperm count' were power transformed (powerTransform function in the 'car' package ver. 3.1-3, Fox et al. 2024) to meet the assumption of multivariate normality. In each model, we ran four MCMC chains (each with 5000 iterations and a warmup of 1000) and confirmed convergence ($R_{\text{hat}} = 1$). The effective sample size for each parameter always exceeded 1480, with all samples generated by iterations ($n = 16\ 000$) retained.

To test if trait variation was affected by the developmental environment, we modelled both the mean and the SD on a logarithmic scale for each trait (and trait covariances) as a function of environment. From the posterior distribution, we obtained the estimated trait mean and SD for each environment. We used the natural logarithm of the ratio between SDs from two environments (i.e. $\ln VR = \ln [SD_1/SD_2]$) to assess the environmental effect on raw variation of each trait. In addition, we used the natural logarithm of the ratio between coefficients of variation (i.e. $CV = SD/\text{mean}$) from two environments (i.e. $\ln CVR = \ln [CV_1/CV_2]$) to assess the environmental effect on relative trait variation while accounting for expected increases in variation with the mean. Notably, $\ln CVR$ assumes a linear relationship between $\ln(\text{mean})$ and $\ln(\text{SD})$, with a slope of 1 (Volkmann et al. 2020, Usui et al. 2021). $\ln CVR$ is particularly useful in when comparing or combining values across studies with a large absolute difference in mean and SD (e.g. meta-analyses; Nakagawa et al. 2015, Bishop et al. 2022). In our study, however, we compared variation of the same trait, where mean values were similar in the three environments. The coefficient of correlation (Pearson's r) between $\ln(\text{mean})$ and $\ln(\text{SD})$ was -0.06 to 0.04 for freshwater, -0.07 to 0.06 for stable salinity and -0.07 to 0.09 for fluctuating salinity among traits in both

sexes and ages. Nevertheless, we reported $\ln CVR$ as a sensitivity analysis to assess relative trait variation. Hereafter, $\ln VR$ refers to raw variation and $\ln CVR$ to relative variation. In most case (45 of 48 tests), the results using $\ln VR$ and $\ln CVR$ were the same. We compared specific pairs of environments to test the effect of salinity (stable salinity / freshwater) and the effect of fluctuations (fluctuating / stable salinity). A positive $\ln VR$ (or $\ln CVR$) indicates an increase in trait variation with salinity or fluctuations in salinity respectively, and vice versa for negative values. The $\ln VR$ (or $\ln CVR$) was significant when its 95% credible intervals excluded zero. We calculated the probability (pMCMC) of the observed effect (null hypothesis = no effect) using the posterior distribution from our models.

Results

Salinity-induced changes in raw trait variation ($\ln VR$)

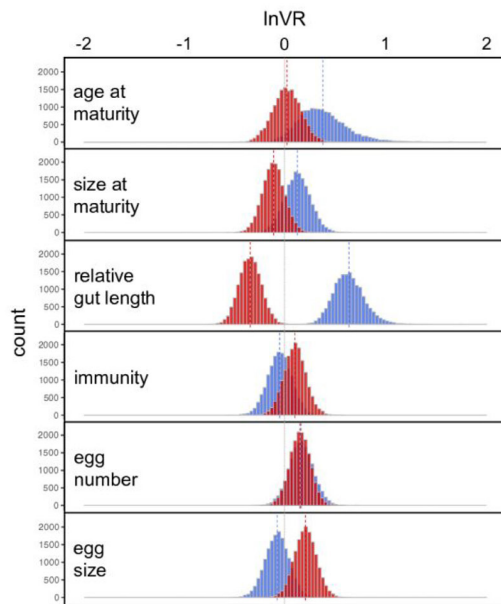
The distribution of raw variation for each trait is shown in Fig. 2. For the 13 traits in young fish (for both sexes), higher but stable salinity significantly affected the raw variation in two traits, while fluctuations in salinity significantly affected the raw variation in three traits (Fig. 3a–b). For the 11 traits in older fish (Fig. 3c–d), however, there were fewer effects of stable salinity or fluctuations in salinity on the raw trait variation: no significant differences in the raw trait variation due to stable salinity, and only one trait showing changes in the raw variation due to fluctuations in salinity. The raw trait variation had the same propensity to be affected by the environment in males (3 of 13 traits) and females (1 of 11 traits). Together, more traits (4 of total 24 traits; Fig. 3) than expected by chance alone (1.2 of 24 traits, if $\alpha = 0.05$) showed a significant difference in the raw variation when reared in different environments.

Female relative gut length when young (+88%) and male age at maturity (+41%) were significantly more variable for fish that developed in water with stable saline rather than freshwater. Young females that developed in water with fluctuating rather than stable salinity had less raw variation in relative gut length (−30%). For males, age and size at maturity (−34%, −35%) were significantly less variable when they developed in water with fluctuating rather than stable salinity. Neither salinity nor its fluctuations affected raw trait variation for older females, but older males that developed in water with fluctuating rather than stable salinity varied significantly less in body size (−25%) (Fig. 3).

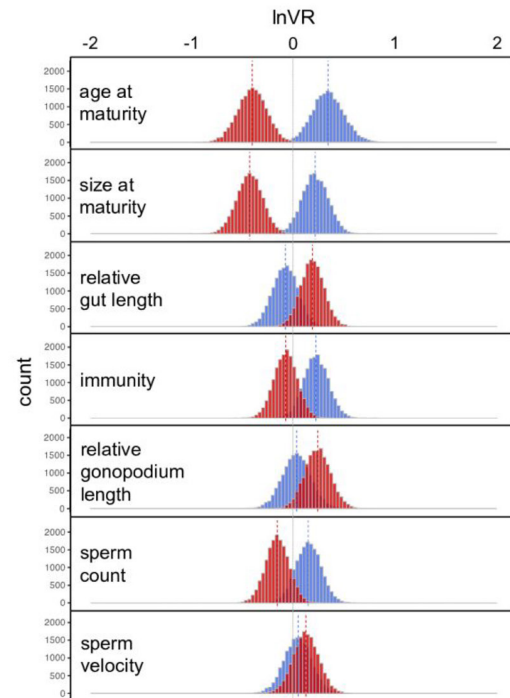
Salinity-induced changes in relative trait variation ($\ln CVR$)

We used $\ln CVR$ to account for trait means in each environment and then test for any changes in relative trait variation (Fig. 4). For the 13 traits in young fish (both sexes), higher but stable salinity significantly affected the relative variation in two traits, while fluctuations in salinity significantly affected the relative variation in four traits (Fig. 5a–b). For the 11 traits in older fish, however, there were fewer effects of

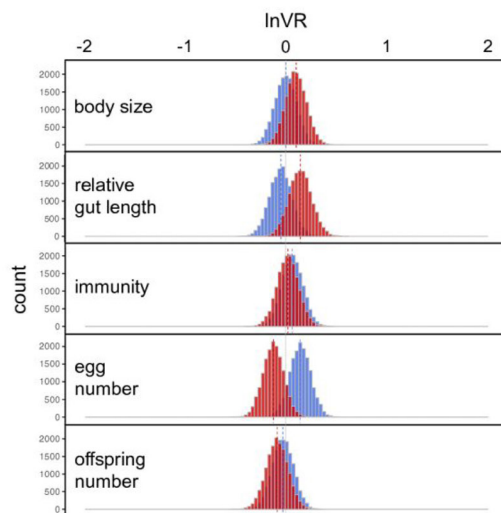
(a) young females



(b) young males



(c) old females



(d) old males

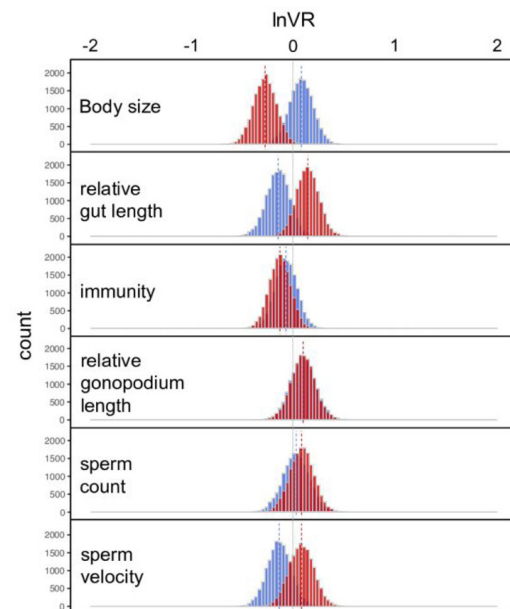


Figure 2. lnVR estimated using posterior distributions showing environmental effects on SD for each trait in (a) young females, (b) young males, (c) old females and (d) old males. Young=newly matured, virgin individuals. Old=older, mated individuals (after 12 weeks of mating). Blue=salinity effect ($\ln SD_{\text{stable salinity}} - \ln SD_{\text{freshwater}}$). Red=fluctuation effect ($\ln SD_{\text{fluctuating salinity}} - \ln SD_{\text{stable salinity}}$). Dash lines represent the mean of lnVR.

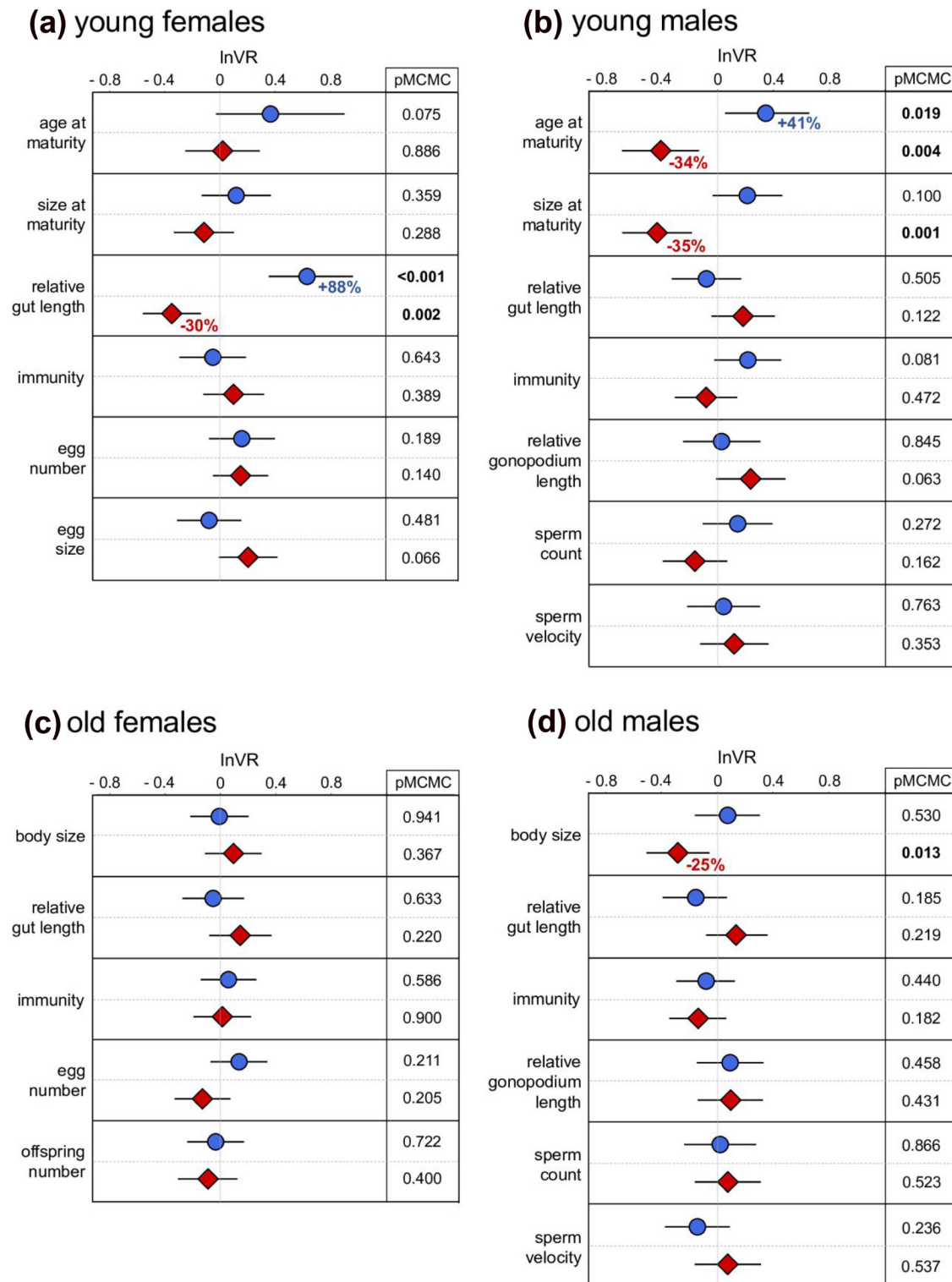
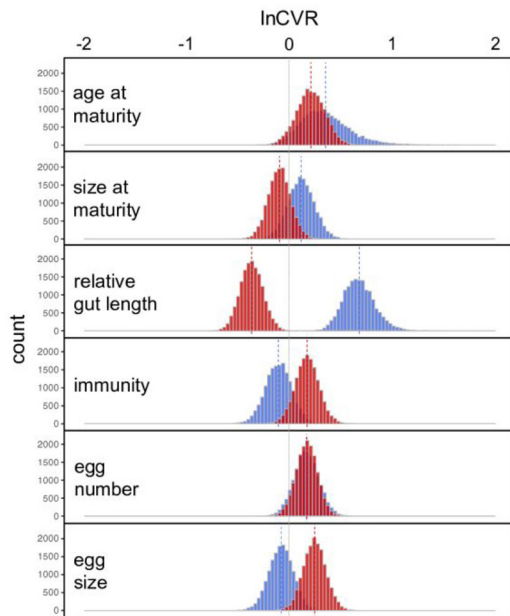
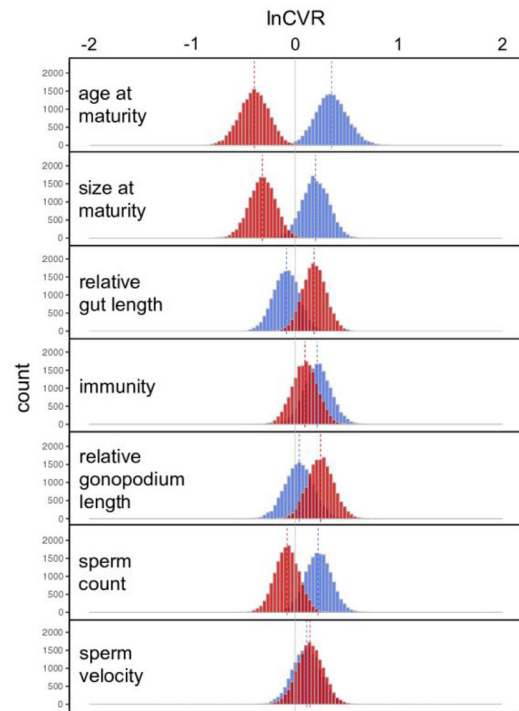


Figure 3. InVR with 95% credible intervals for each trait in (a) young females, (b) young males, (c) old females and (d) old males. Young=newly matured, virgin individuals. Old=older, mated individuals. Blue circle= $\ln\text{SD}_{\text{stable salinity}} - \ln\text{SD}_{\text{freshwater}}$. Red diamond= $\ln\text{SD}_{\text{fluctuating salinity}} - \ln\text{SD}_{\text{stable salinity}}$. Positive values indicate that greater salinity (blue) or fluctuation in salinity (red) 'increases' trait variation, while negative values indicate the opposite. Significant differences in trait variation ($p\text{MCMC} < 0.05$) are shown with the percentage difference, calculated as $[\exp(\text{InVR}) - 1] \times 100\%$.

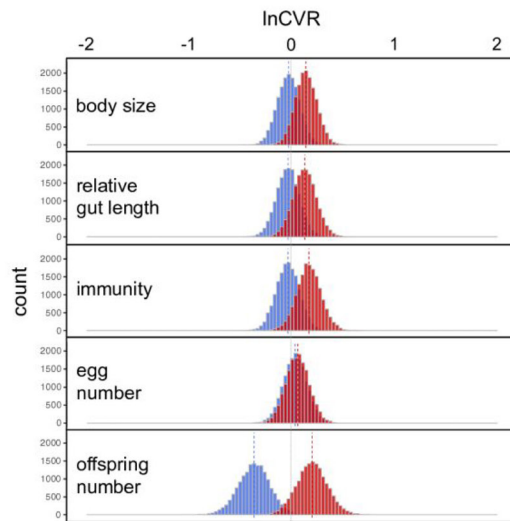
(a) young females



(b) young males



(c) old females



(d) old males

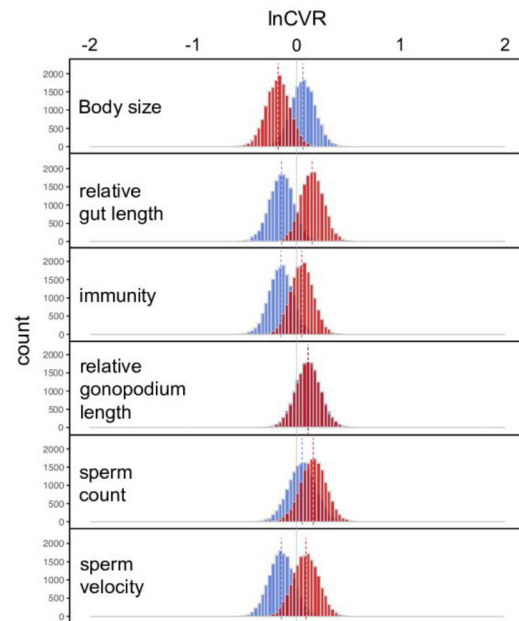


Figure 4. lnCVR estimated using posterior distributions showing environmental effects on CV for each trait in (a) young females, (b) young males, (c) old females and (d) old males. Young = newly matured, virgin individuals. Old = older, mated individuals. Blue = salinity effect ($\ln CV_{\text{stable salinity}} - \ln CV_{\text{freshwater}}$). Red = fluctuation effect ($\ln CV_{\text{fluctuating salinity}} - \ln CV_{\text{stable salinity}}$). Dash lines represent the mean of lnCVR.

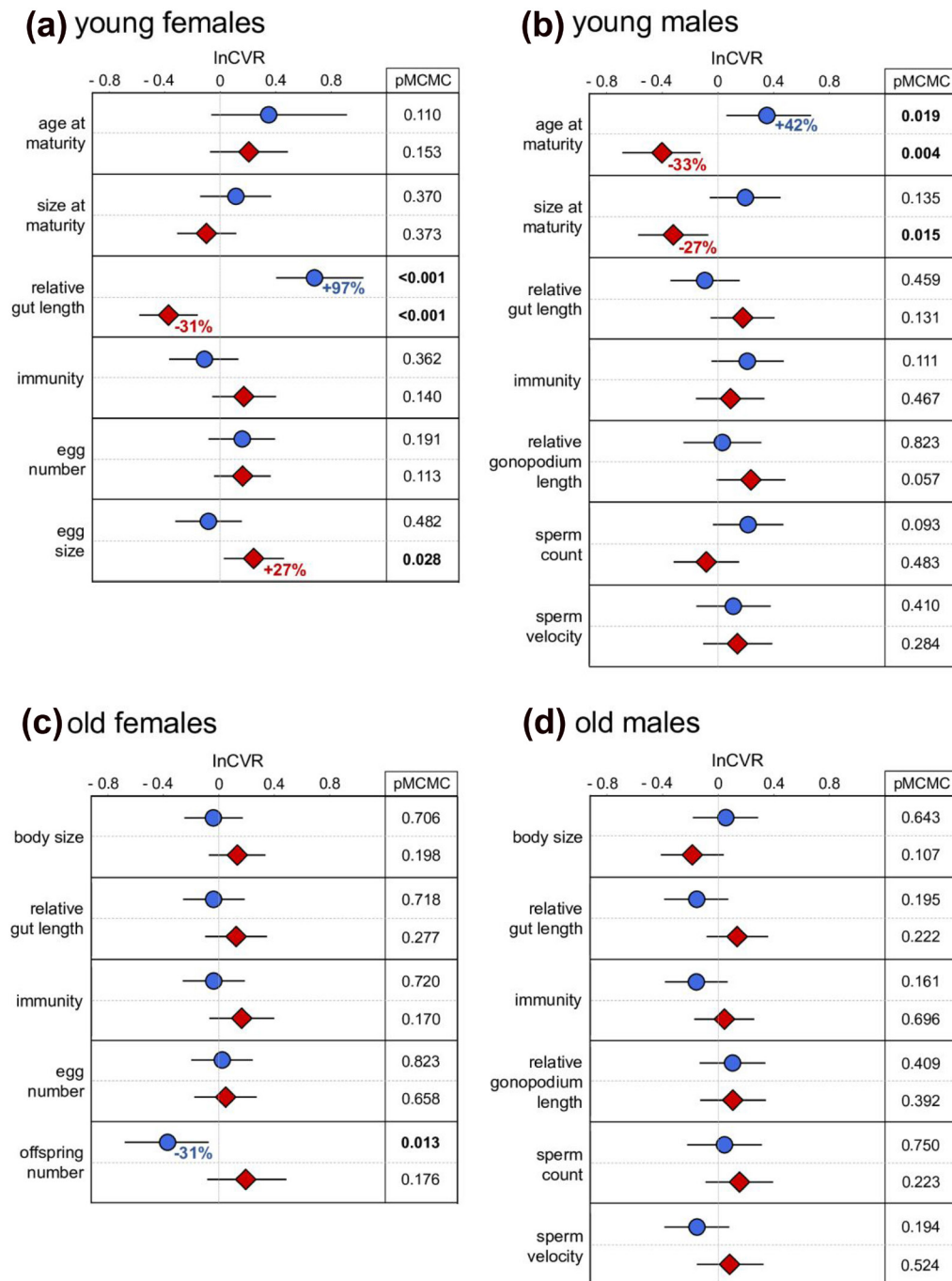


Figure 5. InCVR with 95% credible intervals for each trait in young females (a) and males (b); and old females (c) and males (d). Young=newly matured, virgin individuals. Old=older, mated individuals. Blue circle=salinity effect ($\ln CV_{\text{stable salinity}} - \ln CV_{\text{freshwater}}$). Red diamond=fluctuation effect ($\ln CV_{\text{fluctuating salinity}} - \ln CV_{\text{stable salinity}}$). Positive values indicate that higher salinity (blue) or fluctuation in salinity (red) 'increase' relative variance, while negative values indicate the opposite. Significant differences in relative variance (pMCMC < 0.05) are shown with the percentage difference, calculated as $[\exp(\ln CV) - 1] \times 100\%$.

stable salinity or fluctuations in salinity on the relative trait variation (Fig. 5c–d): only one trait showing changes in its relative variation due to stable salinity, and there were no significant differences in relative trait variation due to fluctuations in salinity. The relative trait variation had the same propensity to be affected by the environment in males (2 of

13 traits) and females (3 of 11 traits). Together, more traits (5 of total 24 traits; Fig. 5) than expected by chance alone (1.2 of 24 traits, if $\alpha = 0.05$) showed a significant difference in the relative variation when reared in different environments.

The environmental effects on InCVR (Fig. 5) were largely consistent with those based on InVR (Fig. 3) for 45 of the

48 tests (i.e. two tests for each of the 24 traits). The exceptions were egg size in young females, offspring number in old females, and body size in old males. Fluctuations in salinity (fluctuating versus stable salinity) significantly increased the relative variation in egg size in young females (+27%), while higher but stable salinity (stable salinity versus freshwater) decreased the relative variation in offspring number in old females (−31%). There was no significant difference in the relative variation in body size between older males from the fluctuating and stable salinity environments (Fig. 5).

Discussion

Variation among individuals experiencing the same environmental conditions often reflect disruption of adaptive responses – such as variation caused by deleterious mutation (Bao et al. 2022) or developmental noise (Saito et al. 2024). Because local conditions typically favour specific trait values, the deviation from those optimal values is detrimental. Variation could, however, also stem from differences in selective pressures (e.g. disruptive, frequency-dependent and bet-hedging), phenotypic plasticity (Fox et al. 2011), gene flow (Yeaman and Jarvis 2006), recombination (Darch et al. 2015) or stochastic variation among individuals that determines optimal allocation to different life history traits. For example, it is well established that sexually selected traits show high phenotypic variation because they tend to be condition-dependent (Dougherty 2021), and males vary in their luck or genetically influenced ability to accrue resources (Snyder and Ellner 2018).

Stable salinity increases, while fluctuating salinity reduces, variation in relative gut length in females but not males

In our study, there were several important changes in trait variation when mosquitofish, *G. holbrooki*, developed under stable or fluctuating salinity when compared to freshwater conditions. Importantly, these changes were always sex-specific. Shortly after maturation, higher salinity (stable salinity versus freshwater) significantly increased (both raw and relative) variation in female relative gut length, while fluctuations in salinity (fluctuating versus stable salinity) reduced both raw and relative variation. No such changes occurred for males. Greater variation in relative gut length at higher salinity might have a negative effect on female osmoregulation (Takei 2021). It might also affect the ability of individual females to utilise different food sources by affecting their capacities for food retention (Karachle and Stergiou 2010). For example, individuals with a relatively longer gut can feed on lower-quality or less digestible food than those with a shorter gut (Olsson et al. 2007). *Gambusia* consume a wide variety of prey with diverse nutrient values (e.g. zooplankton, algae, mosquito larva and tadpoles) (Margaritora et al. 2001, Blanco et al. 2004, Pyke 2005, Remon et al. 2016, Vannini et al. 2018). Greater variation in relative gut length in stable saline environments may

amplify differences in digestive ability between females, resulting in different prey selection or changes in foraging behaviours and/or locations (Araújo et al. 2011, Sánchez-Hernández et al. 2019). This might promote the exploration of new niches with novel prey targets (Araújo et al. 2011). Importantly, *Gambusia* is among the top 100 most invasive species worldwide (Lowe et al. 2000), and predation by *Gambusia* has contributed to the local extinction of endemic fish species (Howe et al. 1997). Female *Gambusia* are typically larger and more aggressive predators than males and juveniles (Blanco et al. 2004). Future research should test if higher salinity, and its fluctuations, generate dietary shifts in females that affect competition with other fish species. Another future challenge is to explain why variation in male relative gut length was unaffected by higher salinity (stable salinity versus freshwater).

Stable salinity increases, while fluctuating salinity reduces, variation in the onset of reproduction in males but not females

Salinity treatments differed between the sexes in how they affected both raw and relative variation in the age and size at maturity. Age and size at maturity were less variable for males from the fluctuating than stable salinity environment, while age (but not size) at maturity was more variable for males from the stable salinity than freshwater environment. No such effects occurred for females. Mating competition among males is intense in *G. holbrooki*, with size-dependent mating strategies (Pilastro et al. 1997). Larger males typically monopolise females by chasing away smaller rivals (Pilastro et al. 1997). Less variation in male age and size at maturity is likely to increase male–male competition: a greater number of similarly sized males will mature at the same time, and male–male contests will take longer to resolve because body size is a major determinant of fight outcome (Harrison et al. 2018). As males barely grow after maturation, fluctuations in salinity also reduced the variation in body size of older males. If a fluctuating salinity environment increases male–male competition, this might benefit females. Male aggression towards other males reduces the time available to harass females, which in turn enhance females' foraging efficiency (Pilastro et al. 2003). In addition, males tend to senesce more rapidly when involved in mating activities (Chung et al. 2024a), and females take longer than males to mature in a fluctuating salinity environment (Chung et al. 2024b). Consequently, when salinity fluctuates, the adult sex ratio may become more female-biased later in the breeding season. This could further increase female fecundity if the benefits of reduced sexual conflict outweigh any costs arising from a lack of suitable mates. In contrast, Alcaraz and García-Berthou (2007) observed a higher proportion of males in wild populations along increasing salinity gradients. Their results suggest that other mechanisms may influence sex ratio in nature. In our source population, newly mature males that were born in spring begin participating in reproductive events in autumn before the breeding season ends in winter (Kahn et al. 2013). The emergence of new males

from the next generation might rebalance the female-biased population that we expected in a fluctuating-salinity environment. Future research should monitor how generations overlap and compare seasonal changes in sexual conflict and female reproduction between environments with stable and fluctuating salinity.

Both stable and fluctuating salinity alter reproductive trait variation in females but not males

Although salinity did not affect raw variation (lnVR) in reproductive traits in either sex, there were sex-specific effects of salinity on relative variation (lnCVR) when we accounted for trait means. The difference between lnVR and lnCVR indicates that the mean and SD differ in how they respond to freshwater salinisation. For example, if a salinity environment increases the mean value but not the SD, this can lead to a negative lnCVR as the variation relative to the mean decreases – a stabilisation effect on trait expression. In our study, fluctuations in salinity led to greater relative variation in egg size, while higher salinity reduced relative variation in how many offspring females produced. There were no equivalent effects on male sperm traits. The greater relative variation in egg size was driven by the reduction in average egg size in a fluctuating salinity environment (Chung et al. 2024b). This finding of smaller eggs corresponds to studies of wild *Gambusia* in Europe showing a reduction in embryo fat contents along increased salinity gradients (Santi et al. 2020). While Santi et al. (2020) reported moderate heritability of embryo fat content – suggesting that a shift in offspring investment can be driven by genetic variation between saline environments – our study suggests such shift can be driven by the environment itself (e.g. phenotypic plasticity). Smaller eggs may increase vulnerability to external stresses (Houde 1987) as a larger surface-to-volume ratio is likely to elevate the demands of osmoregulation (Alderdice 1988). Moreover, greater relative variation in egg size may heighten differences in offspring relative competitiveness (Einum and Fleming 2004, Lalonde 2005, Boivin and Gauvin 2009). Consequently, a proportion of the offspring not only become more susceptible to external environmental pressures but also face competition from larger conspecifics. Our findings suggest a likely short-term outcome in the next generation when females produce eggs that vary more in size. However, for salinity to drive evolutionary changes in offspring investment (Santi et al. 2020), the environment-induced changes in egg production must be heritable (i.e. a genotype-by-environment interaction), and selection must favour these specific changes under fluctuating salinity.

Lower relative variation in offspring number was driven by more offspring being produced by females that developed in the stable salinity than freshwater environment (Chung et al. 2024b). Lower relative differences in female fecundity have several implications. First, a more equal reproductive contribution of females is likely to maintain a higher level of genetic diversity (Miller et al. 2009). Second, there is less opportunity for net selection on females if our

count of offspring is a good proxy for total fitness. This should slow the rate of local adaptation that might improve the ability to survive in a more saline environment. The source population in our study has resided in a freshwater stream for at least 20 years (Chung et al. unpubl.). Although unlikely, we cannot rule out the possibility of historical exposure to higher salinity due to past aridity and human activities in Australia (Cañedo-Argüelles et al. 2016, Thorslund et al. 2021), which could have led to local adaptation to elevated salinity levels. To determine how relative or absolute differences in fecundity affect net selection on females, future studies could compare the genetic diversity of *G. holbrooki* populations from high and low salinity environments.

Conclusion

Salinity treatments altered variation in several reproductive and life-history traits in mosquitofish. In a stable salinity environment, females exhibited greater variation in relative gut length, which might widen their digestive abilities and modify resource competition patterns with other species. Alcaraz et al. (2008) found that *G. holbrooki* exposed to stable salinity took longer to forage and captured fewer prey when they competed with other fish species. Future studies could explore the link between their results and salinity-induced changes in gut development. However, any resultant changes are likely to be reversed under fluctuating salinity because of the reduced variation in relative gut length. On the other hand, a cohort of size-matched males is expected to mature more synchronously in a fluctuating salinity environment due to reduced variation in age and size at maturity. The reduced variation in sexually selected traits in males (in this case: male size) contrasts with the assumption that males exhibit greater variation than females in mating success (Wyman and Rowe 2014). More uniform expression of sexually selected traits can intensify male–male competition (Bookmythe et al. 2013) and drive females to use traits other than body size for mate choice (Heuschele et al. 2009). Human activities and climate change are currently increasing salinity fluctuations in numerous freshwater habitats (Nielsen et al. 2003, Rahel and Olden 2008, Kuan et al. 2012, Dugan et al. 2017). Our study underscores that fluctuations in salinity can generate a completely different effect on trait variation than that seen when there is stable salinity. This may lead to distinct patterns of natural and sexual selection that shape adaptation to saline environments.

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Author contributions

Meng-Han Joseph Chung: Conceptualization (equal); Data curation (lead); Formal analysis (equal); Investigation (lead); Methodology (equal); Visualization (lead); Writing – original draft (lead); Writing – review and editing (equal). **Daniel W. A. Noble:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Validation (equal); Writing – review and editing (equal). **Michael D. Jennions:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Resources (lead); Supervision (equal); Validation (equal); Writing – review and editing (equal).

Data availability statement

Data and code can be downloaded from Mendeley Data (<https://data.mendeley.com/datasets/n7z9nw4rwj/1>) (Chung et al. 2025c).

Supporting information

The Supporting information associated with this article is available with the online version.

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